

## TANAGRA, Greece

WMO: 166990

Lat: 38.3353N

Lon: 23.5628E

Elev: 138

StdP: 99.68

Time Zone: 2.00 (EUE)

Period: 97-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |     | Humidification DP/MCDB and HR |     |      |      |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-----|-------------------------------|-----|------|------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |     | 99.6%                         |     |      | 99%  |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99% | DP                            | HR  | MCDB | DP   | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c) | (d)                           | (e) | (f)  | (g)  | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -1.2       | 0.1 | -6.8                          | 2.2 | 2.8  | -4.2 | 2.7 | 3.2  | 10.3                  | 11.5 | 9.0 | 11.4 | 2.0                   | 300  | 0.420 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.6                   | 36.1            | 20.4 | 34.8 | 20.1 | 33.1 | 19.9 | 22.5                | 31.9 | 21.5 | 30.9 | 20.9 | 30.1 | 4.1                  | 70   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 19.6                            | 14.6 | 25.6 | 18.8 | 13.9 | 24.7 | 17.9 | 13.1 | 24.1 | 66.4          | 31.7 | 63.3 | 30.9 | 61.1 | 30.1 | 26.7              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 8.4           | 7.3  | 6.4 | -4.7                       | 40.1 | 3.2                | 1.5 | -7.0                                               | 41.2 | -8.8       | 42.0 | -10.6      | 42.9 | -12.9      | 44.0 |    |    |
| (5)               |      |     | -5.0                       | 23.7 | 2.9                | 1.5 | -7.1                                               | 24.8 | -8.8       | 25.7 | -10.4      | 26.6 | -12.5      | 27.7 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 16.6   | 7.6   | 8.6   | 10.9  | 14.3  | 18.9  | 23.7  | 26.3  | 26.2  | 22.2  | 17.5  | 13.0  | 9.0   |  |  |
|                                                                        | DBStd       | 7.29   | 3.62  | 3.51  | 3.31  | 2.73  | 2.87  | 2.84  | 2.14  | 2.00  | 2.76  | 2.86  | 3.32  | 3.50  |  |  |
|                                                                        | HDD10.0     | 254    | 92    | 61    | 29    | 2     | 0     | 0     | 0     | 0     | 0     | 0     | 9     | 61    |  |  |
|                                                                        | HDD18.3     | 1493   | 333   | 273   | 232   | 122   | 28    | 1     | 0     | 0     | 3     | 48    | 163   | 291   |  |  |
|                                                                        | CDD10.0     | 2651   | 17    | 21    | 56    | 132   | 277   | 410   | 506   | 502   | 367   | 233   | 100   | 29    |  |  |
|                                                                        | CDD18.3     | 848    | 0     | 0     | 1     | 2     | 47    | 161   | 248   | 244   | 120   | 23    | 3     | 1     |  |  |
|                                                                        | CDH23.3     | 9737   | 0     | 0     | 8     | 61    | 538   | 1896  | 3093  | 2859  | 1099  | 173   | 10    | 0     |  |  |
|                                                                        | CDH26.7     | 4460   | 0     | 0     | 3     | 8     | 129   | 846   | 1611  | 1427  | 411   | 25    | 1     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 2.3    | 2.2   | 2.4   | 2.4   | 2.2   | 2.3   | 2.5   | 2.6   | 2.7   | 2.2   | 1.9   | 1.9   | 2.2   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 430    | 56    | 40    | 53    | 27    | 24    | 7     | 8     | 9     | 20    | 39    | 70    | 76    |  |  |
|                                                                        | PrecMax     | 667    | 131   | 104   | 177   | 61    | 53    | 19    | 34    | 27    | 100   | 141   | 149   | 159   |  |  |
|                                                                        | PrecMin     | 205    | 0     | 9     | 2     | 5     | 1     | 0     | 0     | 0     | 0     | 0     | 19    | 7     |  |  |
|                                                                        | PrecStd     | 114    | 41    | 22    | 44    | 16    | 18    | 6     | 11    | 9     | 29    | 38    | 40    | 43    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 18.8  | 20.2  | 23.7  | 27.9  | 32.0  | 38.0  | 39.9  | 37.8  | 35.2  | 29.1  | 24.8  | 20.9  |  |  |
|                                                                        |             | MCWB   | 12.7  | 13.0  | 13.3  | 15.6  | 18.8  | 19.7  | 20.6  | 20.7  | 19.1  | 18.0  | 16.2  | 15.1  |  |  |
|                                                                        | 2%          | DB     | 17.0  | 18.1  | 21.1  | 25.0  | 29.8  | 34.8  | 36.7  | 35.9  | 32.2  | 27.1  | 22.2  | 18.2  |  |  |
|                                                                        |             | MCWB   | 11.6  | 11.7  | 12.9  | 15.1  | 17.6  | 19.6  | 20.7  | 20.4  | 18.9  | 17.7  | 15.6  | 13.2  |  |  |
|                                                                        | 5%          | DB     | 15.8  | 16.3  | 19.2  | 23.0  | 28.0  | 32.9  | 34.8  | 34.1  | 30.8  | 25.1  | 20.8  | 17.0  |  |  |
|                                                                        |             | MCWB   | 11.3  | 11.2  | 12.1  | 14.4  | 16.7  | 19.5  | 20.3  | 20.2  | 18.6  | 17.2  | 15.4  | 12.3  |  |  |
|                                                                        | 10%         | DB     | 14.1  | 15.1  | 17.9  | 21.2  | 26.2  | 31.1  | 33.2  | 32.9  | 29.0  | 23.8  | 19.0  | 15.2  |  |  |
|                                                                        |             | MCWB   | 10.2  | 10.5  | 11.6  | 13.9  | 16.3  | 18.7  | 20.0  | 20.2  | 18.5  | 16.7  | 14.5  | 11.3  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 13.6  | 14.1  | 14.9  | 17.3  | 20.0  | 22.2  | 23.2  | 23.8  | 22.7  | 20.4  | 18.0  | 15.5  |  |  |
|                                                                        |             | MCDB   | 17.4  | 19.3  | 20.5  | 23.6  | 28.8  | 33.7  | 33.4  | 32.1  | 29.0  | 25.1  | 22.0  | 19.6  |  |  |
|                                                                        | 2%          | WB     | 12.5  | 12.6  | 13.7  | 16.0  | 18.8  | 21.0  | 22.3  | 22.6  | 21.1  | 19.2  | 16.7  | 14.0  |  |  |
|                                                                        |             | MCDB   | 16.1  | 16.6  | 19.2  | 22.3  | 27.0  | 30.9  | 32.3  | 32.0  | 27.3  | 23.9  | 20.7  | 17.6  |  |  |
|                                                                        | 5%          | WB     | 11.5  | 11.7  | 12.8  | 15.2  | 17.8  | 20.3  | 21.4  | 21.6  | 20.2  | 18.4  | 15.8  | 13.0  |  |  |
|                                                                        |             | MCDB   | 14.9  | 15.5  | 17.8  | 21.4  | 25.4  | 30.1  | 31.3  | 31.1  | 26.7  | 23.1  | 19.5  | 16.2  |  |  |
|                                                                        | 10%         | WB     | 10.5  | 10.8  | 11.9  | 14.4  | 17.0  | 19.6  | 20.8  | 20.9  | 19.5  | 17.6  | 15.0  | 11.7  |  |  |
|                                                                        |             | MCDB   | 13.4  | 14.3  | 17.0  | 20.5  | 24.3  | 29.1  | 30.7  | 30.1  | 26.2  | 22.1  | 18.3  | 15.0  |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        |        | 8.4   | 9.4   | 10.5  | 12.0  | 12.5  | 12.8  | 12.6  | 12.1  | 11.4  | 10.0  | 9.0   | 8.3   |  |  |
|                                                                        | 5% DB       | MCDBR  | 10.7  | 11.7  | 13.5  | 15.1  | 14.9  | 15.3  | 15.4  | 14.5  | 14.6  | 12.6  | 11.2  | 10.9  |  |  |
|                                                                        |             | MCWBR  | 6.8   | 7.0   | 7.2   | 7.2   | 6.0   | 5.4   | 5.4   | 5.1   | 5.5   | 5.8   | 6.0   | 6.9   |  |  |
|                                                                        | 5% WB       | MCDBR  | 9.9   | 10.8  | 12.2  | 13.4  | 13.4  | 13.8  | 13.1  | 12.9  | 11.9  | 10.8  | 10.0  | 9.3   |  |  |
|                                                                        |             | MCWBR  | 6.7   | 6.9   | 7.1   | 7.2   | 6.0   | 5.5   | 5.3   | 5.4   | 5.4   | 5.6   | 6.0   | 6.4   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.354 | 0.379 | 0.417 | 0.460 | 0.442 | 0.423 | 0.427 | 0.435 | 0.429 | 0.403 | 0.381 | 0.363 |  |  |
|                                                                        | taud        |        | 2.403 | 2.311 | 2.200 | 2.090 | 2.207 | 2.301 | 2.300 | 2.268 | 2.276 | 2.324 | 2.347 | 2.370 |  |  |
|                                                                        | Ebn at Noon |        | 813   | 837   | 838   | 822   | 844   | 858   | 851   | 836   | 816   | 799   | 770   | 769   |  |  |
|                                                                        | Edh at Noon |        | 87    | 108   | 133   | 158   | 143   | 130   | 129   | 131   | 122   | 106   | 92    | 85    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.87  | 2.64  | 3.96  | 5.30  | 6.40  | 7.34  | 7.48  | 6.65  | 4.96  | 3.33  | 2.22  | 1.62  |  |  |
|                                                                        | RadStd      |        | 0.25  | 0.34  | 0.35  | 0.51  | 0.34  | 0.40  | 0.19  | 0.17  | 0.30  | 0.31  | 0.25  | 0.24  |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (11 neighbors) |  | +0.33 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -80     | N/A     | N/A     |

Nomenclature: See separate page